

OVERVIEW

QView3D is an open-source platform designed to streamline management and communication with 3D printers.

- Supports concurrent control of multiple printers
- Provides real-time monitoring and job queue management
- Cross-platform compatibility (Windows, macOS, Linux)

RESEARCH QUESTIONS

- How can we enable seamless management of 3D printer arrays?
- How do we support diverse printer models and firmware's?
- What is required to enable real-time monitoring of 3D printers?
- How can we ensure cross platform compatibility?

TECHNOLOGIES & METHODS

Backend stack

- Flask web framework with SQLite database
- PySerial for USB G-code communication
- WebSockets (Socket.IO) for real-time updates
- Unicorn to deploy the application to production



Frontend stack

- Vue.js Composition API with TypeScript
- TailwindCSS for responsive design
- GCodePreviewer for viewing a virtual render of what the 3D printer is printing



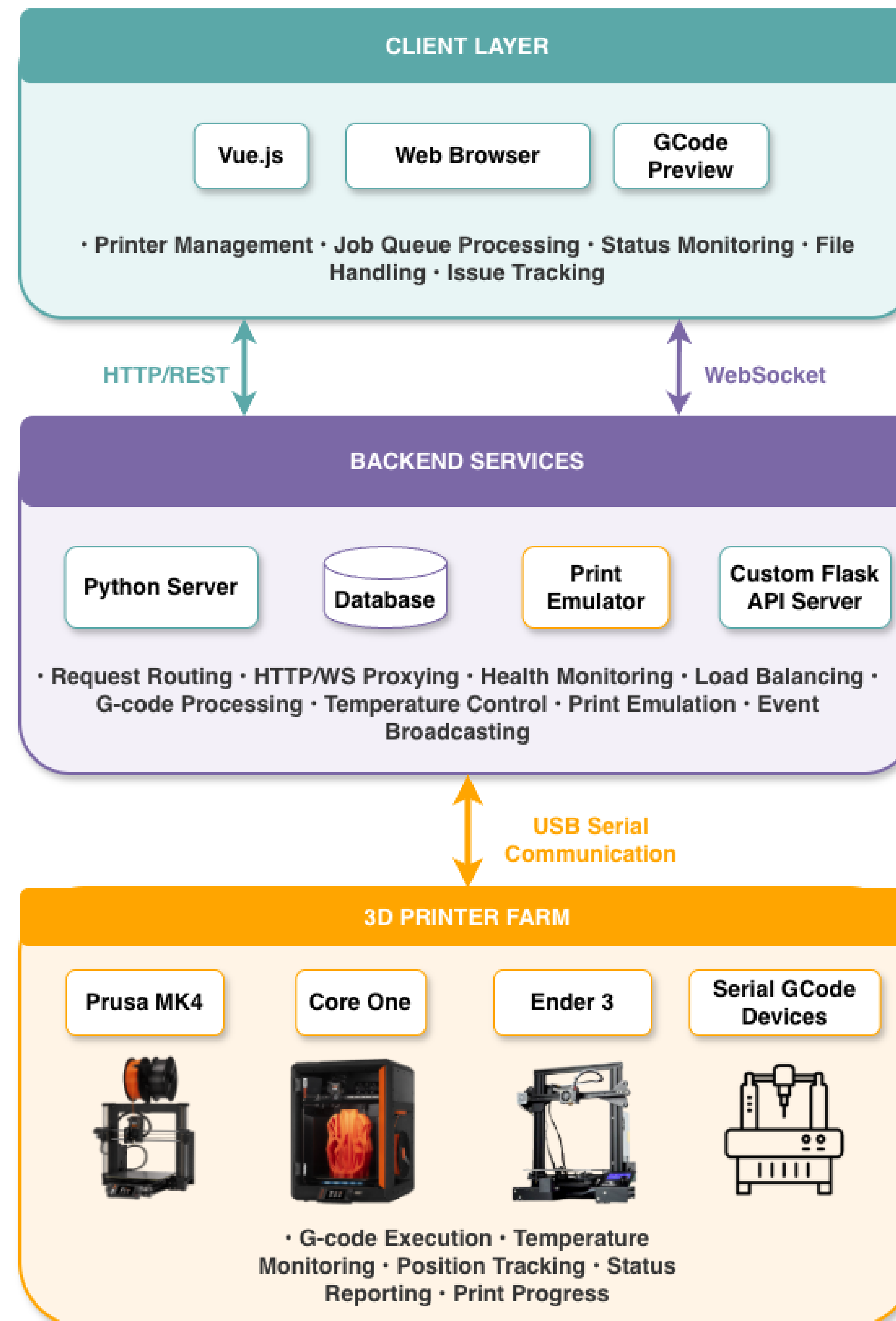
ACKNOWLEDGEMENTS

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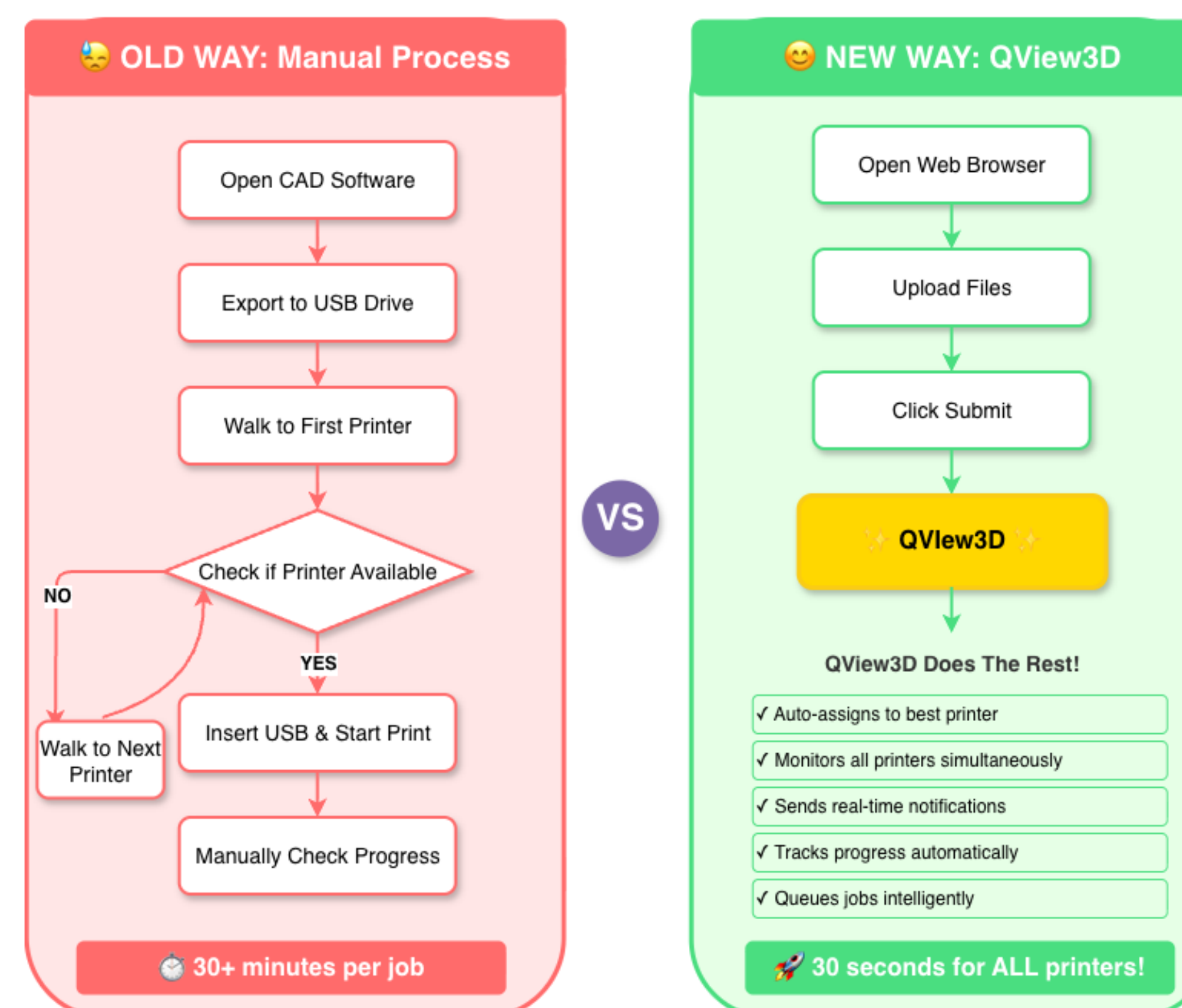
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QView3D System Architecture



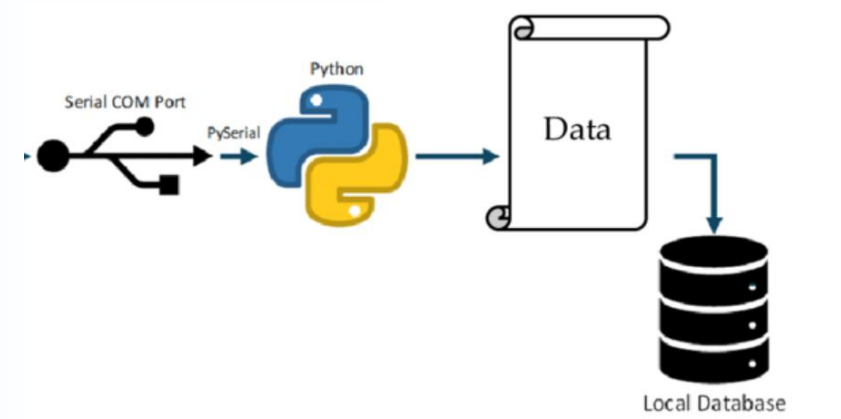
3D Printer Farm Management



RESULTS & ACHIEVEMENTS

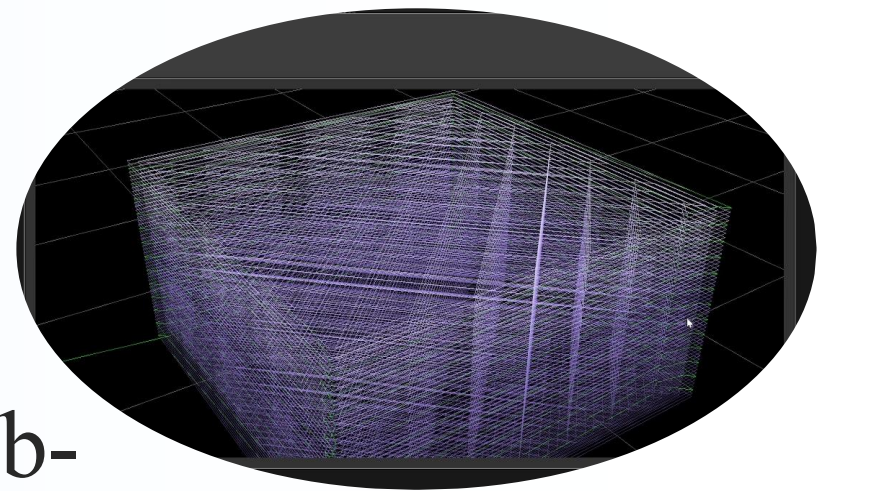
MULTI-PRINTER MANAGEMENT

- Concurrent communication with multiple printers via USB serial
- Automatic job assignment to available printers



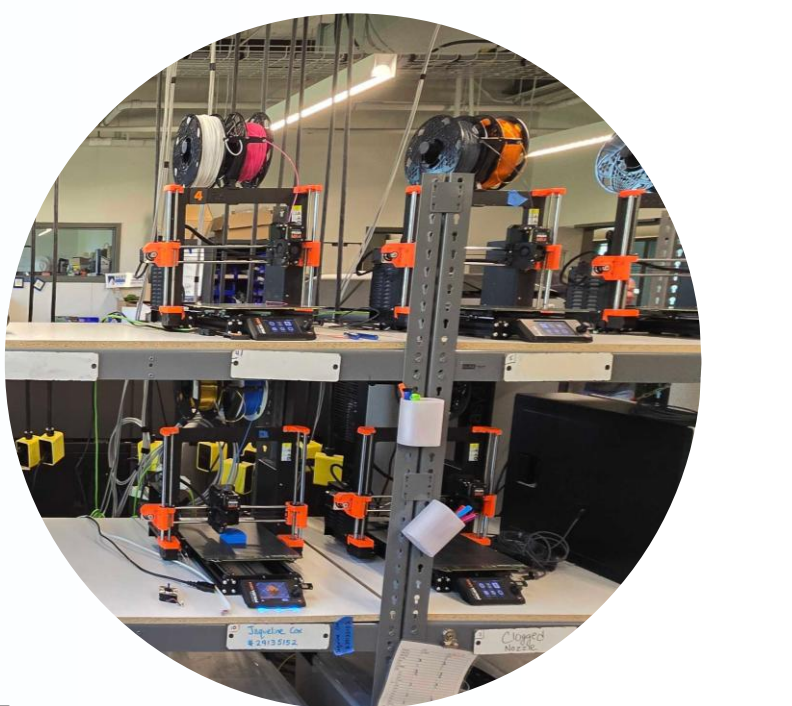
REAL-TIME MONITORING

- Live 3D model preview during printing
- Layer-by-layer G-Code visualization
- WebSocket based status updates with sub-second latency



ADVANCED JOB MANAGEMENT

- Job prioritization and queue management
- Comprehensive job history tracking
- Issue tracking system with error logging
- CSV export for job analytics and reporting



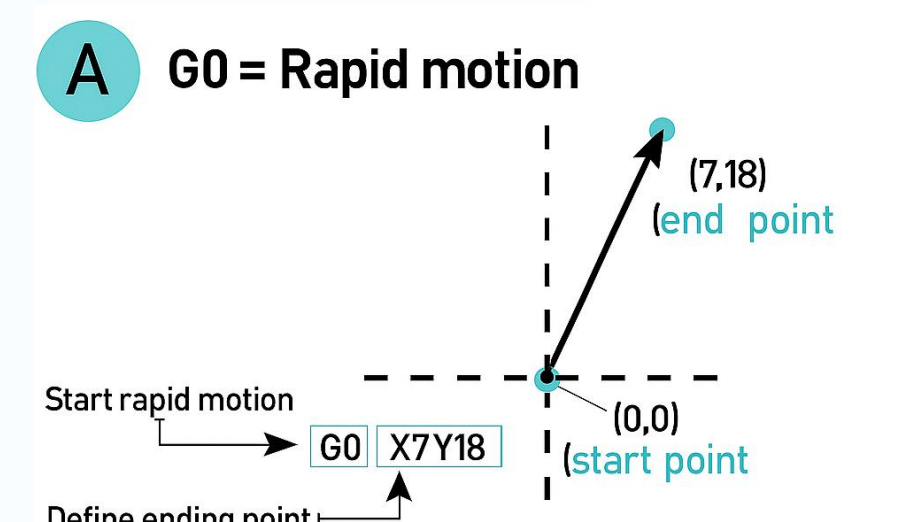
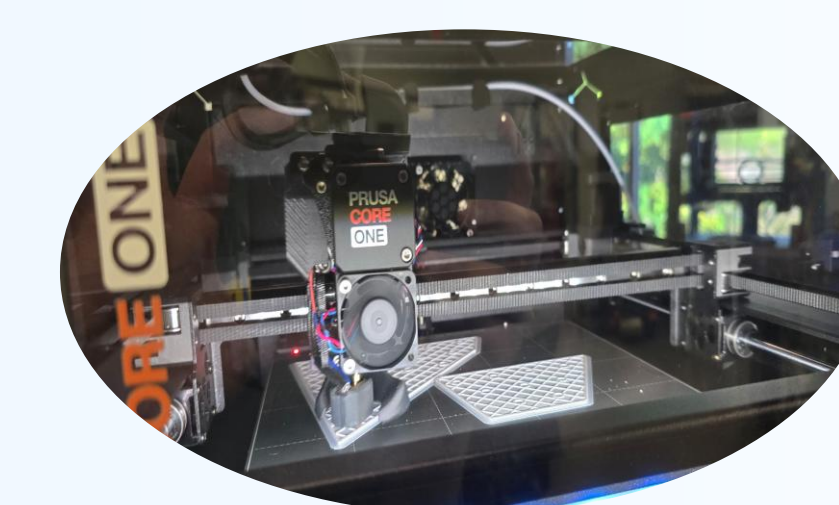
CROSS-PLATFORM SUPPORT

- Support for Windows, Linux and macOS
- Web-based interface accessible from any device



PRINTER MODEL FLEXIBILITY

- Prusa MK3/MK4/MK4S support with custom G-code
- Creality Ender 3/Pro compatibility
- Generic G-code printer support via USB serial
- Expansion to other G-Code devices



REFERENCES

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4. Marlin Firmware. G-code Implementation Reference. marlinfw.org
5. Socket.IO Documentation. Real-time Communication. socket.io



Check out our Github!

github.com/sunhydralab/QView3D

